REG.NO.:

VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Electronics Engineering
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

SLOT:D1+TD1

Programme Name & Branch	**: B.Tech ECE**
Course Code and Course Name	**: BECE204L-Microprocessors and Microcontrollers**
Faculty Name(s)	**:Dr.S.Sundar(Course Coordinator)**
Class Number(s)	**: VL2024250502420/2422/4687/4689/4690**
Date of Examination	**: 30/1/2025**
Exam Duration	**: 90 minutes** **Maximum Marks: 50**

General instruction(s):

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes
  CO3 - Comprehend the architectures and programming of 8051 microcontroller
  CO4 - Deploy the implementation of various peripherals such as general purpose input/ output, timers, serial communication, LCD, keypad and ADC with 8051 microcontroller

Q. No	Question	M	CO	BL
1.	**a). Identify, if the following 8051 instructions have any error.If so give the correct instructions:** (i) MUL A, B (ii) PUSH A (iii) ANL R0,A (iv) MOV A,@R7 (v) MOV R2, R1	5	3	2

	b). Identify the Addressing Mode of the following instructions:
	MOV 03H, 05H
	MOV 03H, #05H
	MOV A, R7
	MOVC A,@A+DPTR
	ADD A,@R0
		5	3	2

2.	Write an assembly language program to add the following BCD numbers together and store the lower byte of the result in R3 and higher byte of the result in R4 . Assume that the numbers are stored in the ROM space starting from 250H onwards. ORG 250H DATA_1: DB 54H,76H,65H,98H,88H,99H Notice that you must first bring the data from ROM space into the RAM and then add them together.	10	3	3
3.	Develop an 8051-assembly language program to do the following task: Assume a switch is connected to port pin P1.0 and monitor its status continuously. If the status of the switch is high, transfer data from Port P0 to Port P2 and blink an LED connected to P3.1. If the switch status is low, transfer data from Port P0 to Port P3 and trigger a buzzer connected to P2.1. Use a delay subroutine for blinking the LED or triggering the buzzer.	10	4	3

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<table>
<tr><td>4.</td><td>Show the output of the following code after the execution and fill the results in Table 1
MOV R0,#30H
MOV 30H,#23
MOV A,@R0
MOV R1,A
PUSH 0
PUSH 1
INC R1
PUSH 2
PUSH 1
POP 3
POP 7
MOV A,R3
PUSH 3
POP 6
END

Table -1
<table><tr><th>Reg/Location</th><th>Value in Hex</th></tr><tr><td>R0</td><td></td></tr><tr><td>R1</td><td></td></tr><tr><td>R3</td><td></td></tr><tr><td>R7</td><td></td></tr><tr><td>R6</td><td></td></tr></table></td><td>10</td><td>3</td><td>3</td></tr>
<tr><td rowspan="2">5.</td><td>(i) Analyse the following assembly code
SETB PSW.4
MOV R2, #78H
MOV A, R2
SETB PSW.3
MOV R4,#0A4H
ADD A,R4
END
(a) Specify the address of the RAM locations, where the data 78H and A4H are stored.
(b) Specify the data stored in accumulator and the status of flags after the addition instruction.</td><td>5</td><td rowspan="2">3</td><td rowspan="2">2</td></tr>
<tr><td>(ii) State the contents of accumulator A in hex, after executing the instruction at each line of the following program. Assume register B has a value of 04H.

ANL A, #00H
MOV A, # 3AH
SWAP A
ANL A, #23H
DIV AB
END</td><td>5</td></tr>
</table>

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